

AMENDMENT NO. 1 JUNE 2026
TO
IS 548 (PART 1/SEC 2) : 2021 METHODS OF SAMPLING AND TEST FOR OILS
AND FATS
PART 1 SAMPLING, PHYSICAL AND CHEMICAL TESTS
SECTION 2 PHYSICAL AND CHEMICAL TESTS

(Page 29, clause **22**) — ‘Delete’.

(Page 33, clause **26**) — Insert the following at the end:

‘27 DETERMINATION OF RANCIDITY– KRIES TEST

27.0 General

Fatty acids, which are integral parts of fats and oils, may have unsaturated carbon-carbon bonds. Some may have a single double bond; some may have multiple. Such double bonds are susceptible to auto-oxidation and the oil becomes rancid. Food items with rancid oils/fats in it become spoilt and unsafe. This can happen in raw oil or refined oils.

Rancidity can be determined qualitatively and quantitatively by kries test. Among the chemical methods, kries test is a promising indicator for detection of rancidity, particularly aldehydes with a characteristic odour impact.

27.1 Principle

Auto-oxidation at fatty acid’s double bonds occurs with molecular oxygen present in the atmosphere, causing the formation of labile peroxides which are unstable and decompose into highly reactive free radicals. These initiate chain reactions which eventually lead to the decomposition of the fatty acid into various low-molecular weight aldehydes and ketones. Aldehydes and ketones react with phloroglucinol developing a pink colour. The colour development in the test is critical and requires optimization.

27.2 Reagents

27.2.1 Phloroglucinol — 0.1 percent solution in diethyl ether

27.2.2 Phloroglucinol — 1 percent solution in glacial acetic acid

27.2.3 Diethyl Ether

27.2.4 Concentrated Hydrochloric Acid

27.2.5 Trichloroacetic acid — 30 percent solution in glacial acetic acid.

27.3 Apparatus

27.3.1 Lovibond Cell — 1 inch

27.3.2 Lovibond Color Glasses — (red yellow and blue)

27.3.3 UV-Vis Spectrophotometer

27.4 Procedure

27.4.1 Qualitative Method

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Shake 5 ml oil vigorously with 5 ml of 0.1 percent phloroglucinol solution in a 15 ml glass test tube (27.2.1). Add 5 ml of concentrated hydrochloric acid. Appearance of pink colour indicates incipient rancidity.

27.4.2 Qualitative Method

27.4.2.1 Method-I

Weigh 0.8 g to 1.02 g of oil or melted fat sample into a 100 ml beaker. Add slowly with stirring 20 ml of freshly prepared 0.1 percent phloroglucinol (27.2.1) until sample dissolves completely. Transfer the solution to a separating funnel, add 10 ml concentrated hydrochloric acid, shake well and allow to separate. Run off acid layer into a 1 inch (2.54 mm) Lovibond cell and match the color using red, yellow and blue glasses. Express result as red Lovibond units.

27.4.2.2 Quantitative—method II

Take 5 ml of oil sample in a stoppered test tube of about 30 ml capacity add 5 ml chloroform and shake well. Add 10 ml of 30 percent solution of trichloroacetic acid (27.2.5), 1 ml of 1 percent solution of phloroglucinol (27.2.2) and mix thoroughly. Incubate the mixture at 45 °C for 15 min. Add 4 ml of ethanol and immediately measure the absorbance at 545 nm and report.

27.5 Inference

27.5.1 Qualitative Analysis

Upto 3 red units indicate incipient rancidity, between 3 units and 8 units indicate the end of induction period and over 8 units indicate definite rancidity.

27.5.2 Quantitative Analysis

Absorbance values below 0.15 indicate no rancidity. Absorbance values greater than 0.2 denote incipient rancidity and absorbance values around 1.0 show that the sample is highly rancid.

28 TOTOX VALUE

28.0 Totox value represents the total oxidative state of fat or oil and directly implicates their rancidity and quality. It is another measure to ascertain oxidative state of oils or fats. Lower the totox value, lower is the oxidative rancidity of the sample.

28.1 Totox value = $(2 \times \text{peroxide value}) + p\text{-anisidine value}$.

NOTE — For determination of peroxide value, see 21 and for determination of *p*-anisidine value, see 23.

29 TEST REPORT

29.1 The test report shall include all the details required for the complete identification of the sample, the sampling method used, date of sample received and tested.

29.2 The test report shall show the method used and the result obtained. The report shall also give all operating conditions not specified in this Indian Standard or regarded as optional, as well as any circumstances that may have influenced the result.'